



What is Wrong with These Experiments?



There is at least one problem with each of the following experiments. What could make them better?

1. Mr. Wilcox is trying out a new beard oil to make his beard grow faster for No Shave November. At the end of November, his beard is 3 inches longer. He concludes that since his beard grew 3 inches, the beard oil must work.

a. What is wrong with this experiment?

We don't know how much his beard would have grown with no oil.
Also he only tested one month.

b. What could be done to improve this experiment?

Measure beard growth for a month without oil ← control group
Measure beard growth for several months ← replication

2. The track coach decided to offer the track athletes two options for daily workouts: a rigorous strength building workout, or a relaxed workout of mostly stretching. After two months, the athletes that chose the strength building workout were 11% faster, while the relaxed workout group was only 3% faster.

a. What is wrong with this experiment?

Athletes got to choose which treatment

b. Describe an appropriate method the coach could use to randomly assign the two workouts to the 120 athletes.

Number the athletes from 1 to 120. Using a random number generator, generate 60 unique numbers between 1 and 120.

The athletes corresponding to these numbers will do the strength workout. The remaining athletes do the relaxed workout.

c. What is the benefit of using random assignment?

We can determine if the strength workout actually works.
(so we can show the strength workout caused faster times)

3. On a recent test day, an AP Stats teacher wanted to determine if smart pills would improve grades. As students walked into class, they were randomly assigned a pill that said either "smart pill" or "sugar pill" on it. The test scores were significantly higher for the group of students that took the smart pill.

a. What is wrong with this experiment?

Students knew which treatment they were getting

b. What could be done to improve this experiment?

no labels on the pills ← single blind (double blind if the teacher doesn't know)

Lesson 2.6 – Designing Experiments

QuickNotes

LT#1 Well-designed experiment

- ① Comparison - 2 or more treatments
- ② Random assignment
- ③ Replication - more than 1 in each treatment group
- ④ Control - keep other variables constant

LT#2 Random assignment

- ① Label ② Randomize ③ Assign
- to show causation

Placebo effect - when a fake treatment (placebo) works

Blinding: when subjects (single blind) and/or experimenters (double blind) don't know about treatments.

LT#3

Check Your Understanding

A group of researchers in Africa find a creative way to protect cattle from lion attacks – they paint eyes on the cows' rears. To determine if this treatment is effective, they randomly assign the cattle to one of three treatments: Eyes on their rear, cross-marks on their rear, or nothing on their rear. After 4 years of roaming the plains, the cows with eyes saw no deaths, the cows with a cross suffered 4 deaths, and the cows with no marks suffered 15 deaths.

- a. Explain why it was important to have a control group that didn't get a mark on their rear.

A control group provides baseline data so we have something to compare to.

- b. Suppose the researchers had 1200 cows. Describe how to randomly assign the cows to the treatments.

Label the cows 1 to 1200. Use a random number generator to get 400 numbers (no repeats). Assign the corresponding cows to get the eyes. Repeat the process to select 400 different cows who get crosses. All remaining cows get no marks.

- c. What is the purpose of randomly assigning treatments in this context?

So we can determine if the treatments cause deaths to decrease.

- d. Create an outline showing a completely randomized design for the experiment.

